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[54] MULTIPLIER CIRCUITS

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[52] **U.S. Cl.** **307/229,** 328/145, 328/160
[51] **Int. Cl.** **G06g 7/12**
[58] **Field of Search** 307/229, 230; 328/145, 160; 235/194

[56] **References Cited**

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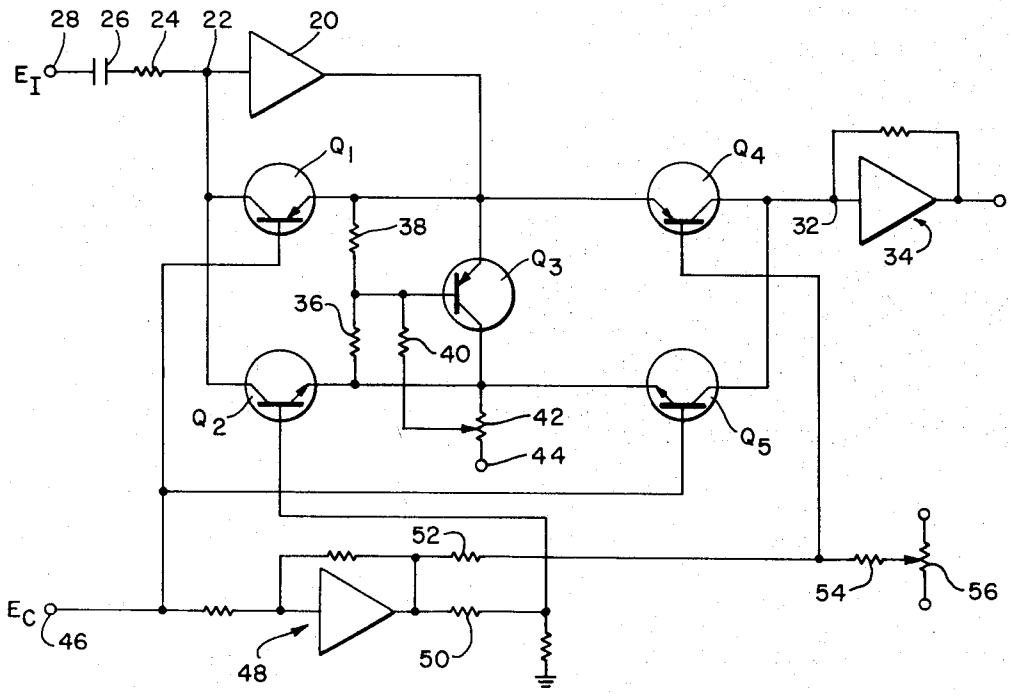
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[57] **ABSTRACT**

A gain control or multiplier circuit in which an input operational amplifier has a pair of feedback paths through respective collector-emitter circuits of opposite conductivity type transistors to form a first bipolar circuit for converting an input signal to a log form by virtue of the log-linear transfer characteristics of the transistors. Each transistor of the first circuit has connected to it another transistor for converting the log signal into its antilog. A second operational amplifier is used as an output buffer for the resulting combined output signals from the antilog transistors. One version employs a bias circuit connected between the emitters of the first bipolar circuit transistors to adjust quiescent current. Another version uses a neutralization circuit to pump currents into the input summing junctions of both operational amplifiers to adjust for capacitive storage effects. In all cases, a control voltage is summed with the log signal by applying the voltage to the bases of the log and antilog converting transistors, thereby controlling the gain between the two operational amplifiers.

14 Claims, 2 Drawing Figures



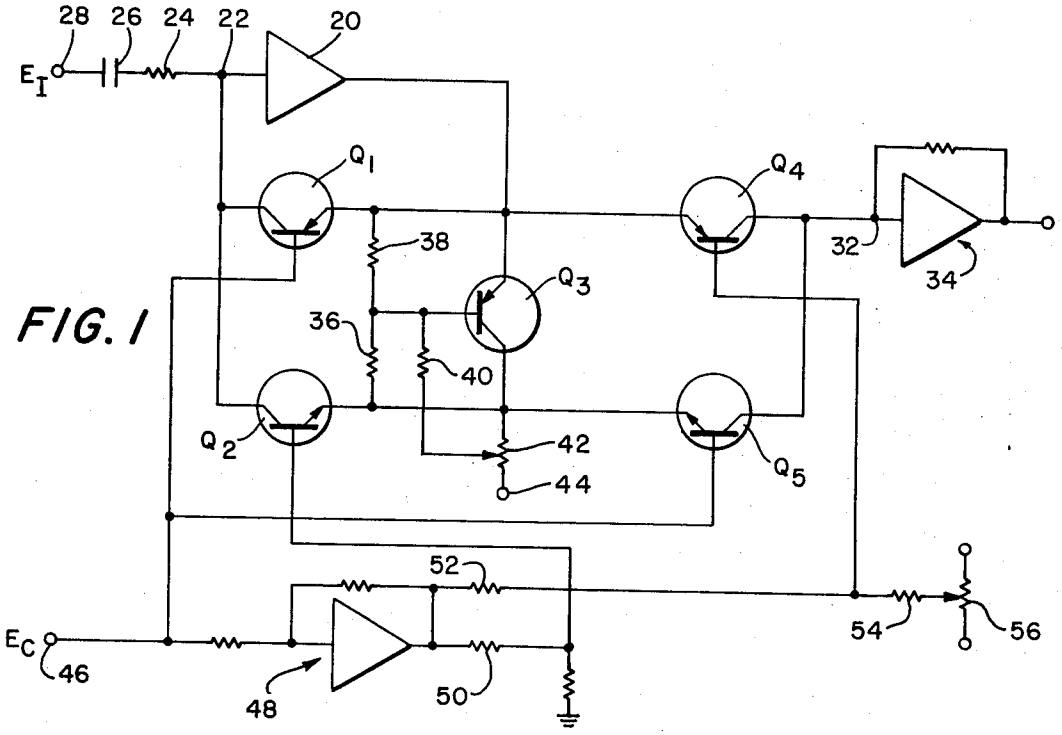


FIG. 1

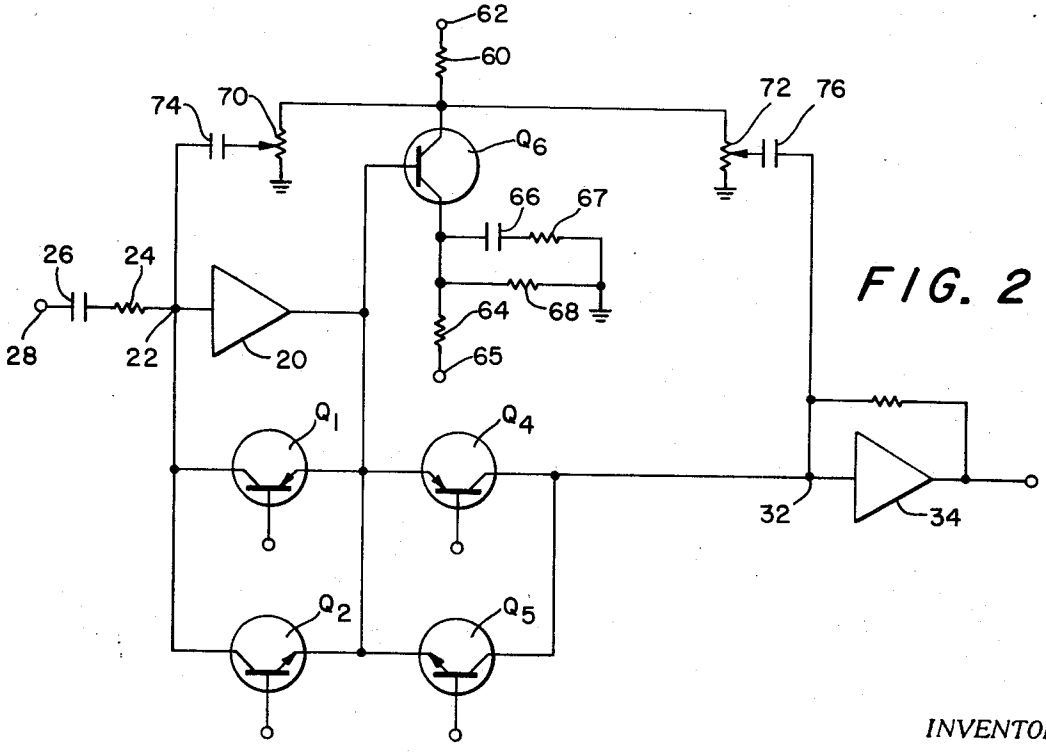


FIG. 2

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MULTIPLIER CIRCUITS

This invention relates to electronic multipliers or gain control systems and more particularly to analog multipliers with logarithmic control response.

Many systems, especially those using audio signals, include signal gain circuits controlled in response to an electrical command signal. Many such gain control circuits in use now are variable loss systems using a light dependent resistor or a field effect transistor as an element in a voltage divider. Variable gain transistor stages of many types are also used, the simplest of which varies gain by changing collector current. But this latter device suffers from a d.c. axis shift in response to the control function. Analog multipliers using balanced semiconductor pairs avoid this problem. However, many such analog multipliers have a semiconductor current higher than the peak current to be handled which seriously degrades the low level noise performance.

A principal object of the present invention is to provide an analog multiplier which has excellent gain control over at least a ± 50 decibel range with very low distortion and noise and a constant decibels per volt control characteristic.

Generally the present invention comprises a first bipolar circuit for generating a first signal which is logarithmically related to an input signal, and a second bipolar circuit for establishing the antilogarithms of the first signal. The gain of at least one of the foregoing circuits is variable in accordance with control signals. To effect the foregoing the device includes an input operational amplifier with opposite polarity feedback paths through oppositely conductive respective semiconductor junctions each exhibiting a log-linear transfer characteristic. A second pair of such semiconductor junction are provided, each connected to derive the antilogarithm of the output of a respective junction of the first pair. Because the semiconductors are transistors, the gain across the junctions is preferably controllable in accordance with a control voltage applied to the bases of selected transistors. The term "gain," as used herein, is intended to include both positive gain or expansion and attenuation or diminution.

The term "bipolar" as used herein is intended to mean a device which is capable of operating on an input signal of either or both polarities.

The invention accordingly comprises the apparatus possessing the construction, combination or elements and arrangement of parts which are exemplified in the following detailed disclosure, and the scope of the application of which will be indicated in the claims.

For a fuller understanding of the nature and objects of the invention, references should be had to the following detailed description taken in connection with the accompanying drawings wherein:

FIG. 1 is a circuit schematic showing details of a multiplier embodying the principles of the present invention; and

FIG. 2 is a circuit schematic showing the details of an alternative version of the device of FIG. 1.

It is known that, in a number of semiconductor devices having a "diode" junction, the voltage across the junction is a function of the logarithm of the forward current through the junction, i.e., the change in forward current with voltage is exponential. This non-

linear behavior of the junction transfer function has been the basis of a number of amplifier circuits having output and input logarithmically related. In the present apparatus, semiconductors with logarithmic transfer functions and exhibiting variable gain transfer characteristics are employed. The latter gain characteristic is preferably substantially linear is decibels per volt.

Referring now to the drawings, there is shown in FIG. 1, apparatus according to the present invention and including an operational amplifier shown generally as including high-gain inverting amplification stage **20** having its input summing junction **22** connected through input resistor **24** and coupling capacitor **26** to first system input terminal **28**. Stage **20** is designed preferably to have a very low input bias current and voltage offset. A first semiconductor device or transistor Q_1 has its emitter connected to the output of stage **20** and its collector to input junction **22**. Transistor Q_1 and Q_2 are of opposite conductivity types. Another transistor Q_3 is connected with its emitter connected to the emitter of transistor Q_1 and its collector connected to the emitter of transistor Q_2.

Another pair of transistors Q_4 and Q_5 are included, the former having its emitter connected to the emitter of transistor Q_1 and the latter having its emitter connected to the emitter of transistor Q_2. The collectors of transistors Q_4 and Q_5 are connected to one another and to the input summing junction **32** of a second operational amplifier **34**.

It will be seen that transistors Q_1 and Q_4 are both PNP type and are preferably matched for V_{be} within 1 mv at 40μa. Transistor Q_2 and Q_5 are NPN type transistors, preferably similarly matched to one another. Because the log-linear transfer characteristic of transistors is a temperature-sensitive function, transistors Q_1, Q_2, Q_4 and Q_5 are all preferably tightly thermally linked as by mounting closely adjacent one another on a common heat sink.

The emitters of transistors Q_1 and Q_4 are connected through a series pair of resistors **36** and **38** to the connected emitters of transistors Q_2 and Q_5. The output of amplifier stage **20** is connected to the emitter of PNP transistor Q_3 and the joined emitters of transistor Q_1 and Q_4. The base of transistor Q_3 is connected to the junction of resistors **36** and **38**. The base of transistor Q_3 is also connected through resistor **40** to an adjustable tap of potentiometer **42**. Potentiometer **42** is connected between the collector of transistor Q_3 and an input terminal **44** at which a negative voltage, e.g., −16 V, can be applied. It will be apparent that when the collector-emitter circuit of transistor Q_3 is conductive, effectively transistors Q_1 and Q_2 each constitute an oppositely-poled conductive feedback path around amplifier **20**.

A second system input or control terminal **46** is provided, connected to the bases of transistors Q_1 and Q_5. Also connected to terminal **46** is an inverting operational amplifier **48**, the output of which is connected through resistor **50** to the base of transistor Q_2 and through resistor **52** to the base of transistor Q_4. The base of the latter transistor is also connected through resistor **54** to adjustable potentiometer **56**.

In operation, a first signal E_1 such as an audio ac is applied at input terminal **28** and a second signal or control voltage is applied to terminal **46** and thus to the

bases of transistors Q_1 and Q_5 directly. The control voltage is also applied in inverted form to the bases of transistors Q_2 and Q_4, scaled, of course, by resistors **50** and **52**. Transistors Q_1 and Q_2 are connected to provided feedback paths around operational amplifier **20**. The latter transistors, being of opposite conductivity types, function as logarithmic converters respectively to convert the positive and negative portions of the input signal to amplifier **20** into logarithmic form. Transistors Q_4 and Q_5 serve as antilog converters which reconvert the signals from transistors Q_1 and Q_2 into linear currents.

The signal applied to the bases of transistors Q_1 and Q_5 and the inverted form of that signal applied to the bases of transistors Q_2 and Q_4 provides the gain control for the current flowing in the collector-emitter circuits of transistors Q_1, Q_2, Q_4, and Q_5. Alternatively, one can simply control the bases of only transistors Q_1 and Q_5 or the bases of only transistors Q_2 and Q_4 if extremes in gain (or attenuation) are not required. Essentially, the application of the control voltage to the transistor bases approximates adding the control voltage to the tied emitters of transistors Q_1, Q_2, Q_4 and Q_5. This is the equivalent of adding the control signal to the log signal. However, it is preferred to add the signal through the transistor bases, thereby avoiding complicating the circuit with additional amplifiers that would be required to feed directly to the tied emitters.

Resistors **36**, **38**, **40** and potentiometer **42** permit the crossover region between polarities to be filled and are normally selected to provide a quiescent collector current in transistors Q_1, Q_2, Q_4 and Q_5 at a value, typically from 0.1 to 1 a, showing sufficient transistor f_t (gain-bandwidth product) to meet desired frequency response requirements. Specifically, resistors **36** and **38** multiply the temperature coefficient of V_{be} of transistor Q_3 to cause the latter, connected collector-emitter across the emitters of transistors Q_1 and Q_2, to track the sum of the V_{be} temperature coefficients of the latter transistors at their quiescent collector currents. The setting of potentiometer **42** and the value selected for resistor **40** allow the desired value of quiescent current to be set. Similarly the selection of resistor **54** and the setting of potentiometer **56** adjusts for transistor offsets, thereby permitting the gain for negative and positive input signals E_i to be made identical.

If the control voltage E_c at terminal **46** is zero, with balanced transistors the circuit gain will be unity. Typically, E_c will have a control constant of -29.8 mv for $+20$ db gain. This constant will have a temperature coefficient of $+0.33\%/C°$ and is proportional to T absolute. A voltage divider having a ratio inversely proportional to T absolute may be used to feed E_c if temperature invarient gain control is desired.

The frequency response of this circuit will be uniform up to well beyond 20 KHz with gains up $+50$ decibels and losses to -50 db. Some slight change in frequency response occurs at greater gains. Equivalent input noise voltage with $+40$ db gain and 20 KHz noise bandwidth will be about 3.4 microvolts rms which is less than 3 db over the noise due alone to an input resistor of about 22 KΩ. Peak input voltage may be in excess of 100 volts when gain is -20 db, hence the input signal-to-noise ratio should be greater than 140 decibels. Output signal-to-noise ratio is lower but this presents no significant restriction on audio system performance inasmuch as a gain control device may be expected by the nature of its use to have a lower output dynamic range requirement.

An alternative embodiment wherein like numerals denote like parts, is shown in FIG. **2**. In the latter the bias circuit provided by resistors **36**, **38** and **40** and potentiometer **42** is eliminated and a neutralization circuit employed instead.

As shown, FIG. **2** includes input terminal **28** coupled through capacitor **26** and resistor **24** to input summing junction **22** of amplifier **20**. A pair of opposite conductivity type transistors Q_1 and Q_2 are each arranged in feedback path between the output of amplifier **20** and summing junction **22**. Antilog transistors Q_4 and Q_5 are coupled to transistors Q_1 and Q_2 and to one another in the same manner as shown in FIG. **1**. For simplicity in exposition, the bases of transistors Q_1, Q_2, Q_3 and Q_4 are all shown in FIG. **2** as connected to respective terminals, but it is to be understood that they are to be considered connected as shown in FIG. **1** for control by the signal imposed on terminal **46**. The connected collectors of transistors Q_4 and Q_5 are coupled to summing junction **32** at the input of second operational amplifier **34**.

As thus described, the circuit of FIG. **2** is very similar to that of FIG. **1** except that in FIG. **2** the emitters of all four transistors Q_1, Q_2, Q_4 and Q_5 are tied together through a direct connection rather than through the collector-emitter circuit of transistor Q_3.

To the foregoing circuit of FIG. **2**, there is added npn transistor Q_6, the base of which is connected to the output of amplifier **20**. The collector of transistor Q_6 is connected through resistor **60** to terminal **62** at which a supply voltage of one polarity can be applied, e.g., $+16$ volts. The emitter of transistor Q_6 is connected through resistor **64** to another terminal **65** at which the opposite polarity supply voltage, e.g., -16 volts, is to be applied. The emitter is also connected through an RC network of series-connected capacitor **66** and resistor **67** both in parallel with resistor **68**, to ground.

The collector of transistor Q_6 is connected also to ground through the respective resistors of two potentiometers **70** and **72**. The adjustable tap of potentiometer **70** is connected through capacitor **74** to input summing junction **22** of operational amplifier **20**; similarly the adjustable tap of potentiometer **72** is connected through capacitor **76** to input summing junction **32** of operational amplifier **34**.

Preferably, transistor Q_6 is an inverting amplifier with a gain of about 1.5. Capacitor **66** and resistors **67** and **68** are frequency compensating components. In operation, it will be seen that transistor Q_6 provides an inversion of the signal from amplifier **20** and a portion of the inverter signal is fed through potentiometers **70** and **72** and their associated capacitors **74** and **76** to the input and output operational amplifier summing points. Potentiometers **70** and **72** should be adjusted so that the currents supplied to the amplifier summing points are equal and opposite to the currents due to the collector-emitter capacitances of transistors Q_1, Q_2, Q_4 and Q_5. The circuit thus may provide signals at the emitters of the latter transistors, which signal rapidly slews through the "dead zone" between positive and negative conduction. Distortion due to the charge

storage effects in the transistors and to capacitive coupling of the signals from transistor Q_6 are thus greatly reduced. For very broadband variable gain circuits one can employ a combination of the neutralization and bias circuits disclosed respectively in FIGS. 2 and 1.

Since certain changes may be made in the above apparatus without departing from the scope of the invention involved, it is intended that all matter contained in the above description or shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

What is claimed is:

1. An electrical gain control system including

a first bipolar circuit for providing first output signal which is logarithmically related to an input signal thereto;

means for summing a gain control signal with said first output signal;

a second bipolar circuit connected to said first circuit for providing a second output signal which is an anti-logarithmic function of the sum of said first output signal and gain control signal.

2. A system as defined in claim 1 wherein said first circuit comprises a first summing operational amplifier having a pair of negative feedback paths, and a first pair of opposite conductivity type semiconductor devices each disposed to control conduction in a respective one of said feedback paths, each of said devices having a log-linear transfer characteristic.

3. A system as defined in claim 2 wherein said semiconductor devices are transistors having substantially matched V_{be} characteristics and said transistors are mounted on a common heat sink.

4. A system as defined in claim 2 wherein said second circuit comprises a second pair of opposite conductivity type semiconductor devices each connected to the output of a respective one of said first pair of semiconductor devices so as to provide an output signal which is a function of the antilogarithm of the output signals from said first pair of semiconductor devices.

5. A system as defined in claim 4 wherein said semiconductor devices of said second pair are transistor mounted on a common heat sink and have substantially matched V_{be} characteristics.

6. A system as defined in claim 4 wherein all of said semiconductor devices are transistors, each pair having substantially matched V_{be} characteristics, and all are mounted on a common heat sink.

7. A system as defined in claim 1 wherein said means for summing said control signal comprises a control signal terminal connected to control gain of said system for input signals of one polarity and means connected to said terminal for providing an inverted form of said control signal and connected to control gain of said system for input signals of opposite polarity.

8. A system as defined in claim 6 wherein said means for applying said control signal comprises a control signal terminal connected to the base of one transistor of at least one of said pairs, and an inverter having its input connected to said control terminal and its output connected to the base of the other transistor of said one of said pairs.

9. A system as defined in claim 6 wherein said means for applying said control signal comprises a control signal terminal connected to the bases of one transistor of said first pair and one transistor of said second pair, and an inverter having its input connected to said control terminal and its output connected to the bases of the other transistors of said first and second pair.

10. A system as defined in claim 4 including a bias control circuit connected to control quiescent currents in said transistors at a value showing sufficient f_t to meet a desired frequency response of said system.

11. A system as defined in claim 10 wherein said bias control circuit comprises a biasing transistor emitter-collector connected between the emitters of the transistors of said first pair, and a biasing network connected to said biasing transistor so that the latter exhibits a temperature coefficient of V_{be} which matches the sum of the V_{be} temperature coefficients of said transistors of said first and second pairs of their quiescent collector current values.

12. A system as defined in claim 4 including a neutralizing circuit for compensating said system for the total effective collector-emitter capacitances of the semiconductor devices of both said first and second bipolar circuits.

13. A system as defined in claim 12 including second summing operational amplifier havings its input connected to the outputs of said second pair of devices, and wherein said neutralizing circuit comprises an inverter having its input connected to the output of said first operational amplifier and its output coupled to the summing junction inputs of both said first and second operational amplifiers.

14. A system as defined in claim 13 wherein said inverter is a transistor having its base connected to the output of said first operational amplifier and its collector connected through respective series connected potentiometers and capacitors to said summing junction inputs.

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